

When calibration is not enough: a reproducible SKY130 design-space study

1 | A FAIR, REUSABLE EXPERIMENT

A small nominal offset does not guarantee a correct decision before the clock deadline. We compare the same stimuli and reveal wrong, unresolved and uncalibratable cases.

Declared search

Nine physical candidates; three training conditions; six selection inputs. Budget: 2x training core energy and 4x gate-area proxy. The chosen design is frozen before a 49-condition-per-circuit evaluation.

What changes in the circuit?

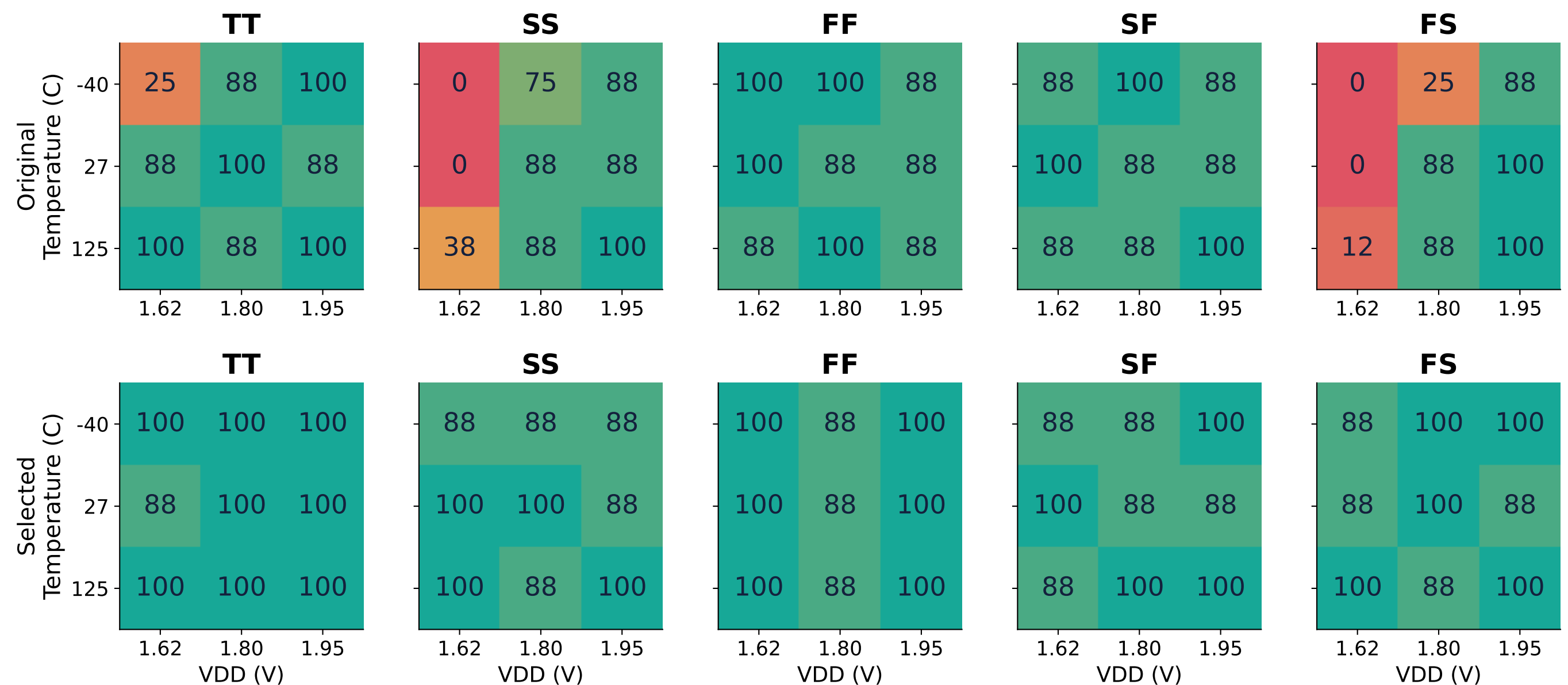
Selected: lvt_balanced_4b
27 transistor instances; sign plus 4 magnitude bits (codes -15 to +15).
Input width: 3 μm ; tail width: 7 μm .
LVT main and auxiliary input devices; standard-VT latch, tail and digital source switches.

What stays honest?

Complementary 80% / 20% output rails are required. Zero input is unscored.
Unavailable calibration remains a failed coverage point, not an invented successful run.
Every figure is backed by a checked notebook, raw SPICE traces and source/model hashes.

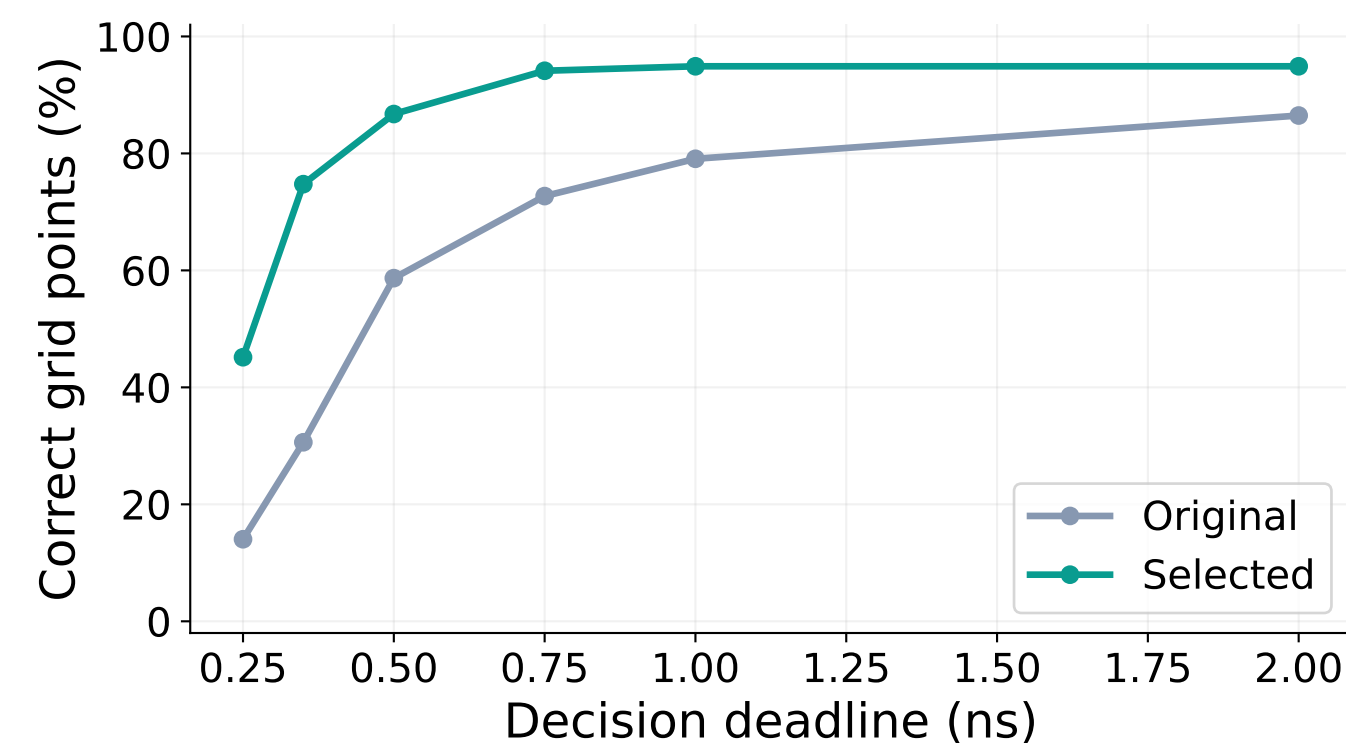
2 | THE FULL PVT GRID, NOT ONE FAVORABLE WAVEFORM

Same local 3.5 ns calibration policy; score at 1 ns; $|\text{input}| \geq 1 \text{ mV}$; +4% width stress



Numbers are passing test-grid percentages, not manufacturing yield. The four extra nominal stress controls are included below.

3 | TIMING, ENERGY AND SCOPE



79.1% -> 94.9%

All 49 conditions; same policy; 1 ns; $|\text{input}| \geq 1 \text{ mV}$

Reserved conditions: 80.2% -> 94.8%
Matched core energy: 143.0 -> 251.5 fJ/cycle
Gate-area proxy: 1.90x original
13,054 distinct physical simulations
996 final numerical deadline checks
14 initial issues; 7 points refined

A design improvement, not a free lunch.

Interface-stress failures remain visible. Schematic simulations only: no silicon, layout/PEX, transient-noise probability or foundry yield. Core energy excludes input/clock drivers and calibration control.

Reproduce: [Comparator_Atlas_Optimized.ipynb](#) | [node scripts\setup.mjs](#) | [offline explorer: results\study\report.html](#)

Established ideas: Razavi, StrongARM Latch (2015), DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka, auxiliary-pair calibration (2022), DOI 10.1007/s10470-022-01992-6.

Original implementation: MIT. SKY130 models retain Apache-2.0. GitHub Copilot assistance disclosed. Authorship supplied by the entrant. No global novelty, IEEE endorsement or award implied.